Bill of Materials - VBF-T6R1NOCEILING-BOM-01

ED-Compact B-p smartphone cell. Dual caliber g/cell and kg/kWh; literature defaults annotated.

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Binder / conductive additive use literature defaults (annotated): the parameter set has no inert-volume parameters; production formulation must rebalance the active fraction. Enclosure and tabs: Not modeled.

Component	g/cell	kg/kWh	Annotation / source
Positive electrode active material (NMC811)	17.7286	0.9531	thickness 75.6 um x area x volume fraction 0.70 x density 3262 kg/m3 (parameter set + params_r2_bp.json)
Positive electrode conductive additive (carbon)	0.3546	0.0191	literature default 2 wt% of coating solid (parameter set has no additive params - annotated); addition requires active-fraction rebalance
Positive electrode binder (PVDF)	0.3546	0.0191	literature default 2 wt% of coating solid (annotated); addition requires active-fraction rebalance
Negative electrode active material (graphite)	11.5991	0.6236	thickness 85.2 um x area x volume fraction 0.80 x density 1657 kg/m3 (parameter set + params_r2_bp.json)
Negative electrode conductive additive (carbon)	0.1160	0.0062	literature default 1 wt% of coating solid (annotated); addition requires active-fraction rebalance
Negative electrode binder (CMC+SBR)	0.2320	0.0125	literature default 2 wt% of coating solid (annotated); addition requires active-fraction rebalance
Separator	0.1945	0.0105	thickness 9 um x area x density 397 kg/m3; porosity 0.47 (parameter set)
Electrolyte	5.4164	0.2912	pore volume 4.514 cm3 x electrolyte density 1.2 g/cm3 (literature value - annotated)
Positive current collector (Al)	2.7729	0.1491	thickness 10 um x area x density 2700 kg/m3 (params_r2_bp.json)
Negative current collector (Cu)	7.3615	0.3958	thickness 8 um x area x density 8960 kg/m3 (params_r2_bp.json)
Enclosure	Not modeled	Not modeled	Not modeled (beyond parameter set) - honest
Tabs	Not modeled	Not modeled	Not modeled (beyond parameter set) - honest
TOTAL (contract caliber, electrolyte excluded)	39.6566	2.1319	electrodes + CCs + separator (calc-energy mass_kg)
Cell energy	0.018601 kWh	-	calc-energy output energy_wh / 1000 (18.6012 Wh)